

CS4092 - Database Design and Development: Final Project

Relational Schema Design: Online Restaurant & Delivery System

1. Standard Relational Schema Notation

(Primary Keys are bolded & underlined; Foreign Keys are italicized with references)

1. Customers (**customer_id**, full_name, email, phone_nb)

2. Addresses (**address_id**, *customer_id*, address_label, street_address, city, state, zip_code)

- Foreign Key: *customer_id* REFERENCES Customers(**customer_id**) ON DELETE CASCADE

3. Staff (**staff_id**, full_name, staff_role)

4. MenuItems (**item_id**, item_name, category, price, is_available)

5. Orders (**order_id**, *customer_id*, *address_id*, *staff_id*, order_date, order_status)

- Foreign Key 1: *customer_id* REFERENCES Customers(**customer_id**) ON DELETE CASCADE

- Foreign Key 2: *address_id* REFERENCES Addresses(**address_id**)

- Foreign Key 3: *staff_id* REFERENCES Staff(**staff_id**) ON DELETE SET NULL

6. OrderItems (**order_item_id**, *order_id*, *item_id*, quantity)

- Foreign Key 1: *order_id* REFERENCES Orders(**order_id**) ON DELETE CASCADE

- Foreign Key 2: *item_id* REFERENCES MenuItems(**item_id**) ON DELETE CASCADE

2. Detailed Table Specifications

Table: Customers

- customer_id: INT, AUTO_INCREMENT, PRIMARY KEY
- full_name: VARCHAR(50), NOT NULL
- email: VARCHAR(50), UNIQUE, NOT NULL

- phone_nb: CHAR(10)

Table: Addresses

- address_id: INT, AUTO_INCREMENT, PRIMARY KEY
- customer_id: INT, NOT NULL, FOREIGN KEY -> Customers(customer_id)
- address_label: VARCHAR(20), DEFAULT 'Home'
- street_address: VARCHAR(100), NOT NULL
- city: VARCHAR(50), NOT NULL
- state: VARCHAR(20), NOT NULL
- zip_code: VARCHAR(10), NOT NULL

Table: Staff

- staff_id: INT, AUTO_INCREMENT, PRIMARY KEY
- full_name: VARCHAR(50), NOT NULL
- staff_role: VARCHAR(50), NOT NULL

Table: MenuItem

- item_id: INT, AUTO_INCREMENT, PRIMARY KEY
- item_name: VARCHAR(50), NOT NULL
- category: VARCHAR(50), NOT NULL
- price: DECIMAL(10,2), NOT NULL
- is_available: BOOLEAN, DEFAULT TRUE

Table: Orders

- order_id: INT, AUTO_INCREMENT, PRIMARY KEY
- customer_id: INT, NOT NULL, FOREIGN KEY -> Customers(customer_id)

- address_id: INT, NOT NULL, FOREIGN KEY -> Addresses(address_id)
- staff_id: INT, NULL, FOREIGN KEY -> Staff(staff_id)
- order_date: TIMESTAMP, DEFAULT CURRENT_TIMESTAMP
- order_status: VARCHAR(20), DEFAULT 'Placed'

Table: OrderItems

- order_item_id: INT, AUTO_INCREMENT, PRIMARY KEY
- order_id: INT, NOT NULL, FOREIGN KEY -> Orders(order_id)
- item_id: INT, NOT NULL, FOREIGN KEY -> MenuItem(item_id)
- quantity: INT, NOT NULL, CHECK (quantity > 0)